

Load Interconnection Request System Impact Study Report

For Queue Project Q23-2343

Blount 100 kV
September 19, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2343. The Interconnection Customer proposes a 28.6 MW large load (data center) facility interconnecting at the Blount 100 kV substation in the Ohio area. The request was received on August 23, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 588 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$276,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Fiddymment – Miami 100 KV L	Queen Creek – Baraga Vlg 500 KV	100.6	184	5.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fiddymment – Miami 100 KV L	Reconductor line	\$1,428,000	19% (\$276,000)
	Total allocated to project		\$276,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.